

Sentence-Level Factuality Classification in Brazilian Portuguese News

An NLP Classification Task Using Fine-Tuned Machine Learning Models

FACTNEWS DATASET

BRAZILIAN PORTUGUESE

TRANSFORMERS

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Task 1 Recap

Feature Representations

- Bag of Words (BoW)
- TF-IDF
- FastText

Data Imbalance Issue

- Bias is 7.6× smaller than Factual
- Bias sentences longest on average
- Factual has widest spread

Models Implemented

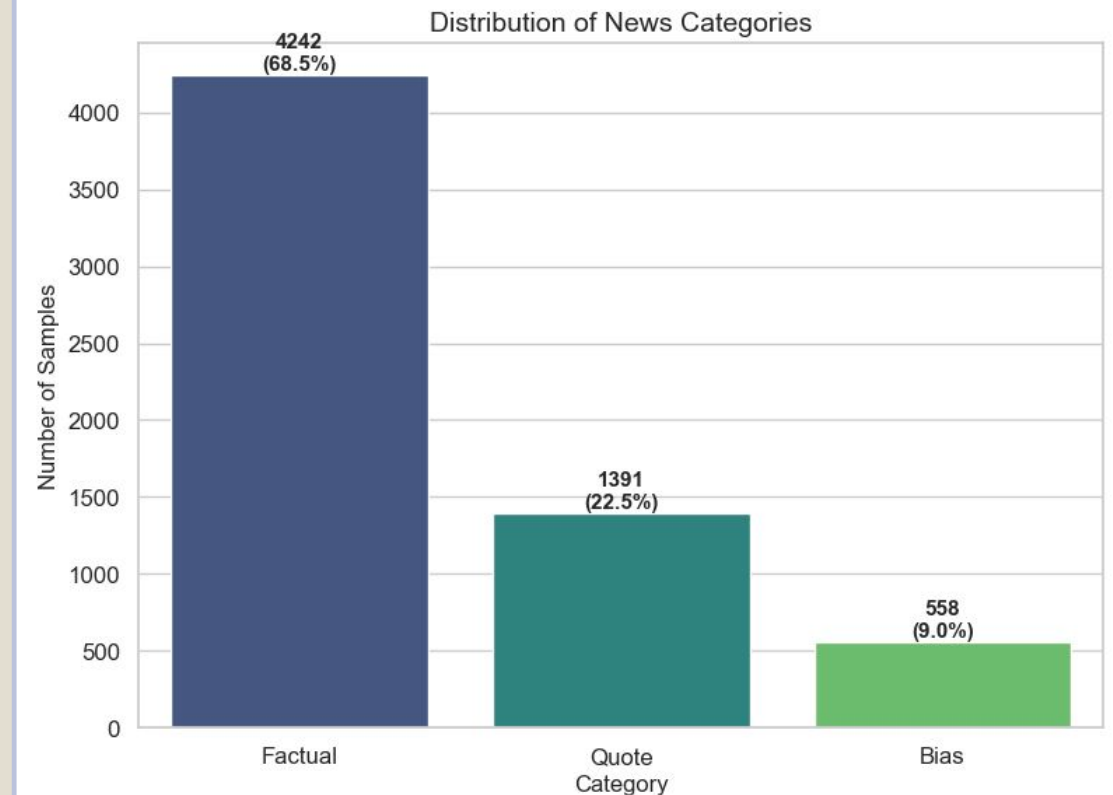
- Decision Trees
- Random Forest
- SVM
- Logistic Regression
- Naive Bayes

Preprocessing Techniques

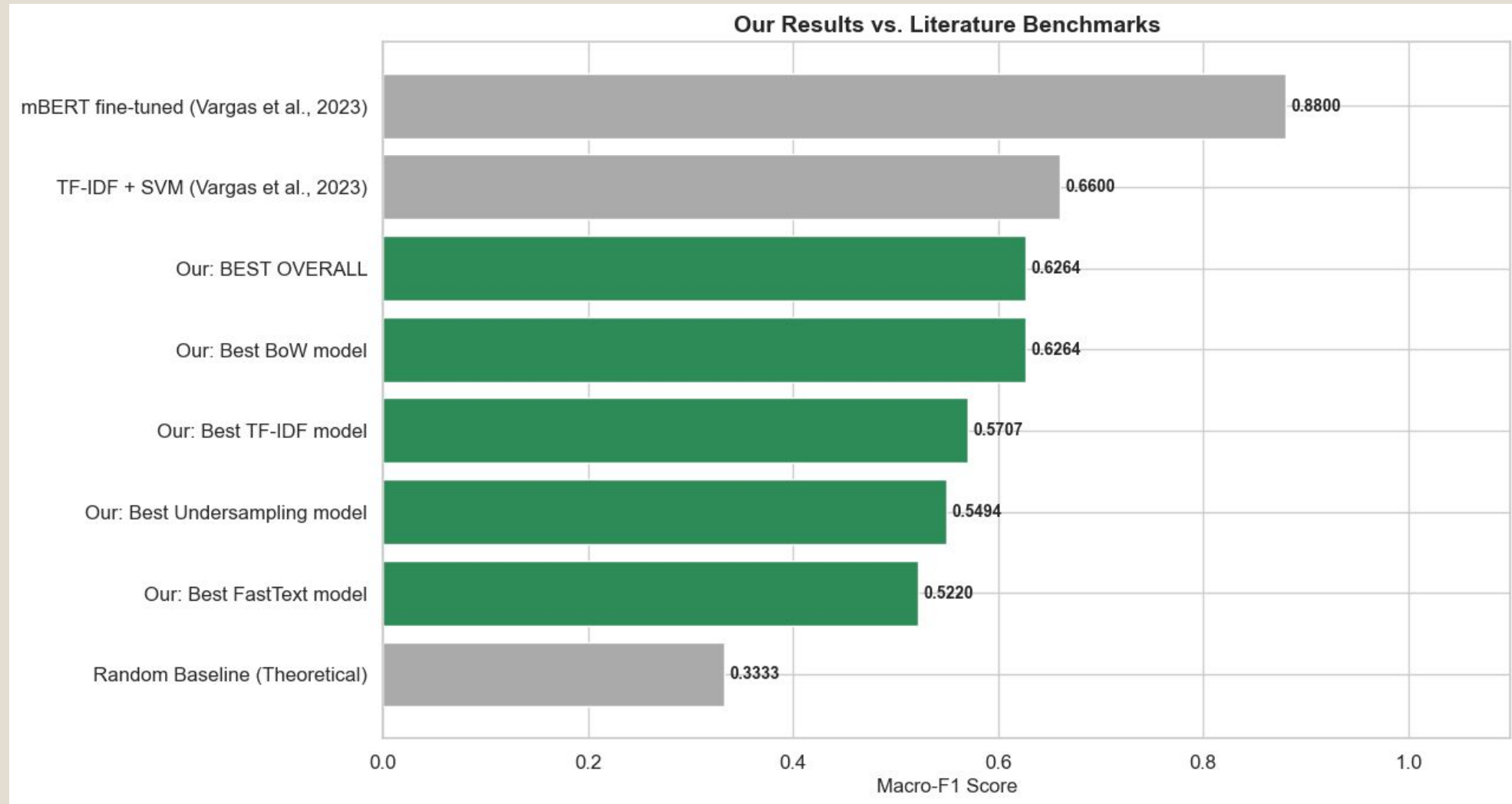
- Lowercasing
- Remove non-alphabetical characters
- Tokenisation
- Stop-word Removal
- RSLP Stemming

Key findings

- Sparse representations (BoW) outperformed dense (FastText) for this task
- Bias VS Factual is the primary confusion pattern across all models
- Class imbalance (Bias = 9%) is the core challenge
- Undersampling improved Bias recall at a small Macro-F1 cost
- Limitations: stemming degrades embeddings, FastText loses sentence structure, no deep learning by design



Task 1 Best Results



NB + BoW (0.626) is the A1 target to beat. The gap to mBERT (0.880) motivates transformer fine-tuning in A2.



Sparse > Dense

BoW/TF-IDF outperformed FastText averaged embeddings



Primary Confusion

Bias ↔ Factual — subtle opinion language resembles factual reporting



Imbalance is Core

7.6:1 ratio; undersampling helped Bias recall marginally

BERTimbau: Primary model

neuralmind/bert-base-portuguese-cased

Model Information

[BERTimbau](#) is a pre-trained BERT model trained on Brazilian Portuguese data.

The BERT-Base Cased variant was Pre-trained on Brazilian Portuguese Wikipedia + BrWaC for 1,000,000 steps, using whole-word masks.

This model achieves state-of-the-art performances on three downstream NLP tasks: **Named Entity Recognition**, **Sentence Textual Similarity** and **Recognizing Textual Entailment**

- Same family as **Vargas et al.**: mBERT
- **Details:** 110M parameters; 12 attention layers

❏ **Key decision:** No RSLP stemming or stop word removal. Transformers use their own subword tokenization (WordPiece / SentencePiece). Raw, unprocessed text is the optimal input.

XLNet-RoBERTa: Secondary model

FacebookAI/xlm-roberta-base

Model Information

Introduced in the paper [Unsupervised Cross-lingual Representation Learning at Scale](#), by Conneau et al., and first released in [this repository](#).

XLNet-RoBERTa is a multilingual version of RoBERTa.

It is a Transformer pre-trained on 2.5TB of filtered *CommonCrawl* data containing 100 languages. It was pre-trained on the raw texts only, with no humans labelling them in any way.

- **Details:** 278M parameters · 12 attention layers

❏ “Note that this model is primarily aimed at being fine-tuned on tasks that use the whole sentence (potentially masked) to make decisions, such as **sequence classification**, token classification or question answering.”

Fine-tuning Configuration & Class Imbalance Handling

Training Hyperparameters

Hyperparameter	Value	Justification
Epochs	10	Small dataset needs more passes
Learning rate	2e-5	Standard BERT fine-tuning LR
Batch size	8 (×2 accum. = 16 eff.)	Fits Kaggle GPU VRAM
Warmup ratio	10%	Prevents early instability
Weight decay	0.01	L2 regularisation
Precision	fp16 (mixed)	GPU memory efficiency

Imbalance Strategies

Weighted Cross-Entropy Loss: Primary

Custom `WeightedTrainer` overrides `compute_loss`

Class weights: Bias **5.9×**, Quote **1.8×**, Factual **0.49×**

Uses full training data: no information discarded

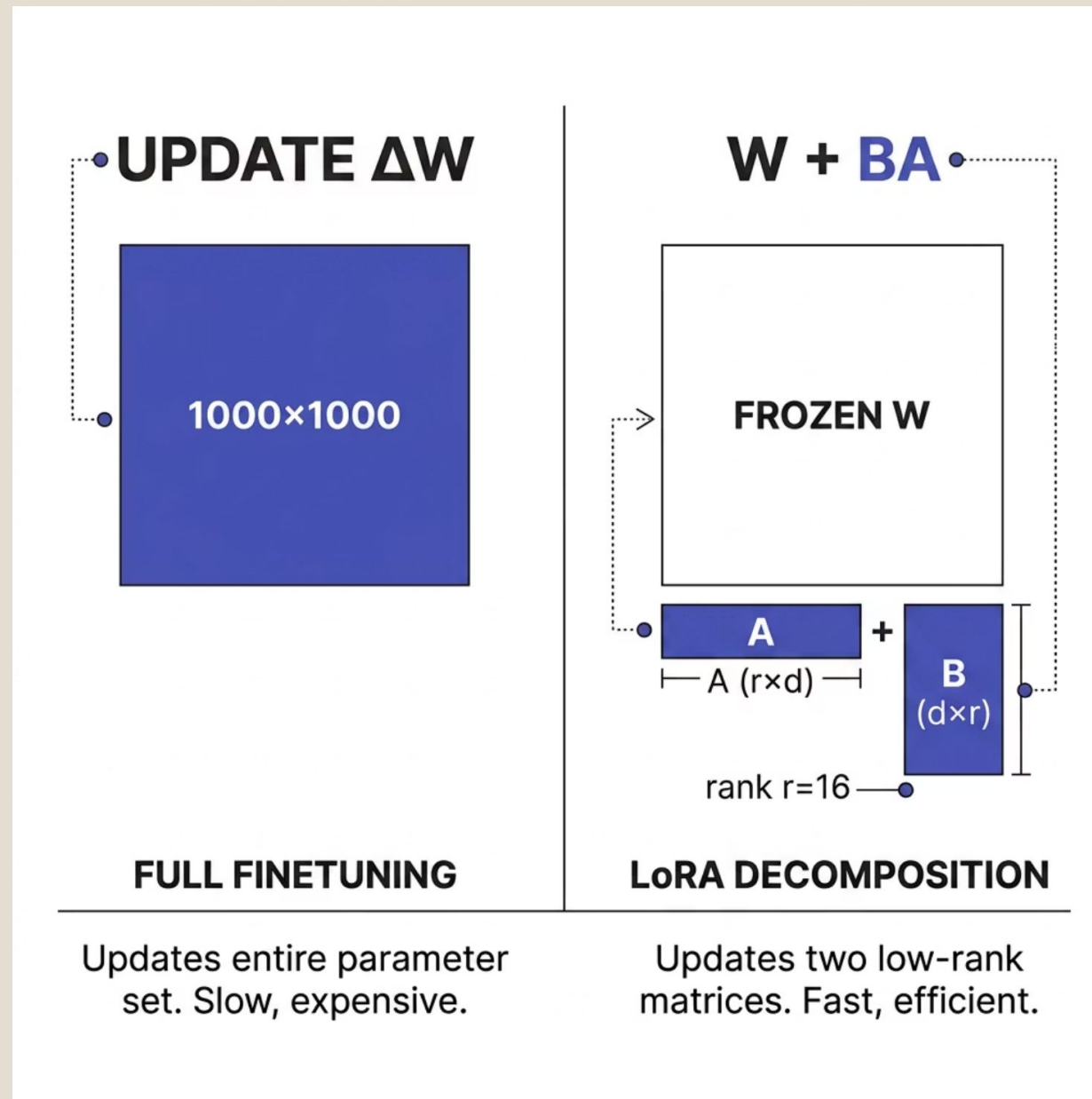
Undersampling: Comparison

All classes reduced to 446 samples (Bias size)

Training set: 4,953 → 1,338 samples

Loses majority-class training signal

Parameter-Efficient Fine-tuning with LoRA (Hu et al., 2022)



Configuration Used

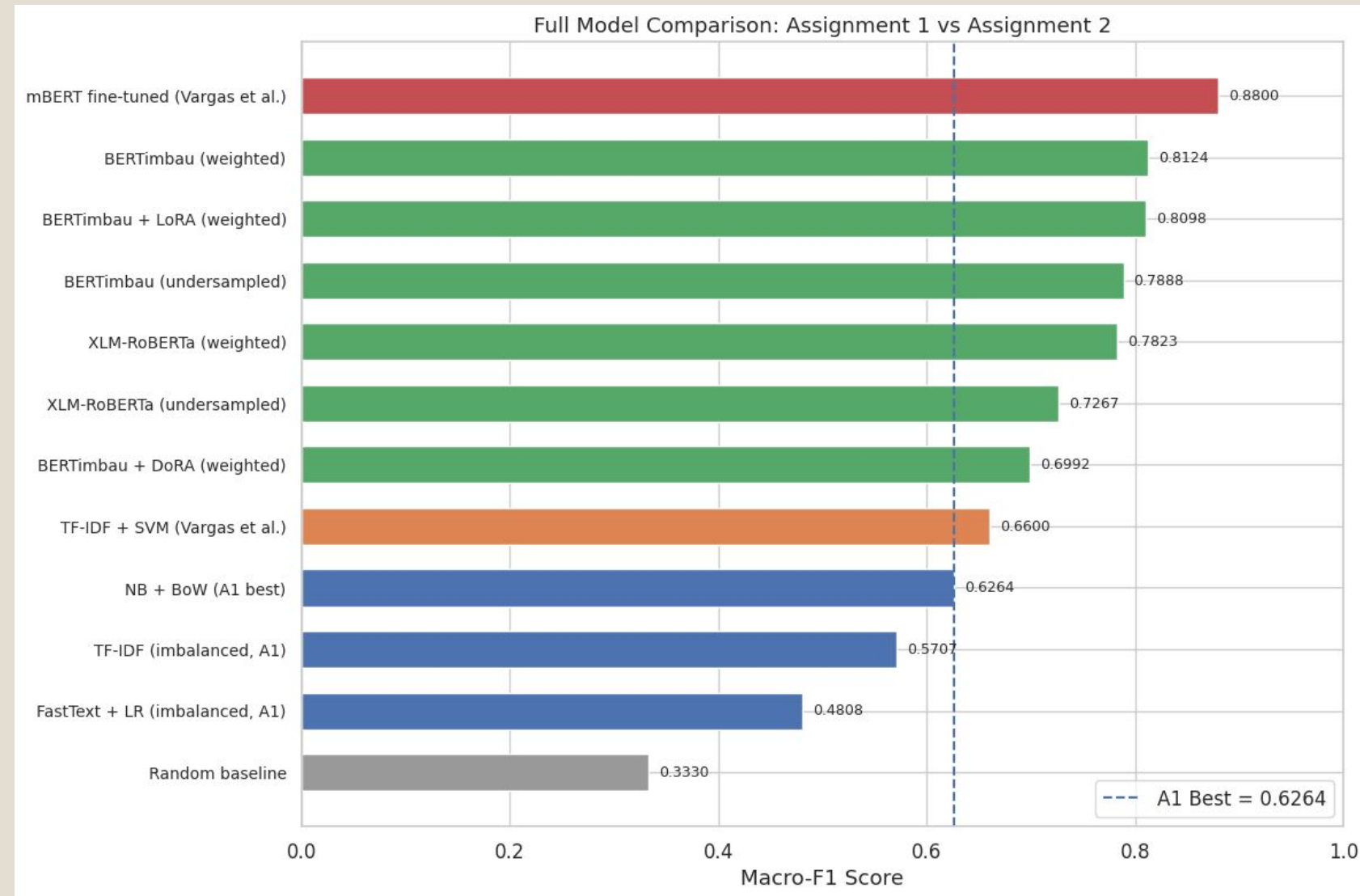
- Rank (r) = 16
- Alpha = 32 (scaling: $\alpha/r = 2$)
- Dropout = 0.1
- Target modules: Query, Key, Value, Dense attention matrices
- Learning rate: $5e-4$ (higher $\rightarrow$ appropriate for adapter layers only)

Parameter Efficiency

Configuration	Params	% of Params
Full fine-tuning	110M	100%
LoRA	2.5M	~2.5%

 Kaggle GPU memory is limited — LoRA fits where full fine-tuning may OOM. Small dataset — pre-trained weights transfer well with minimal adaptation.

Results - 3-Class Factuality Classification

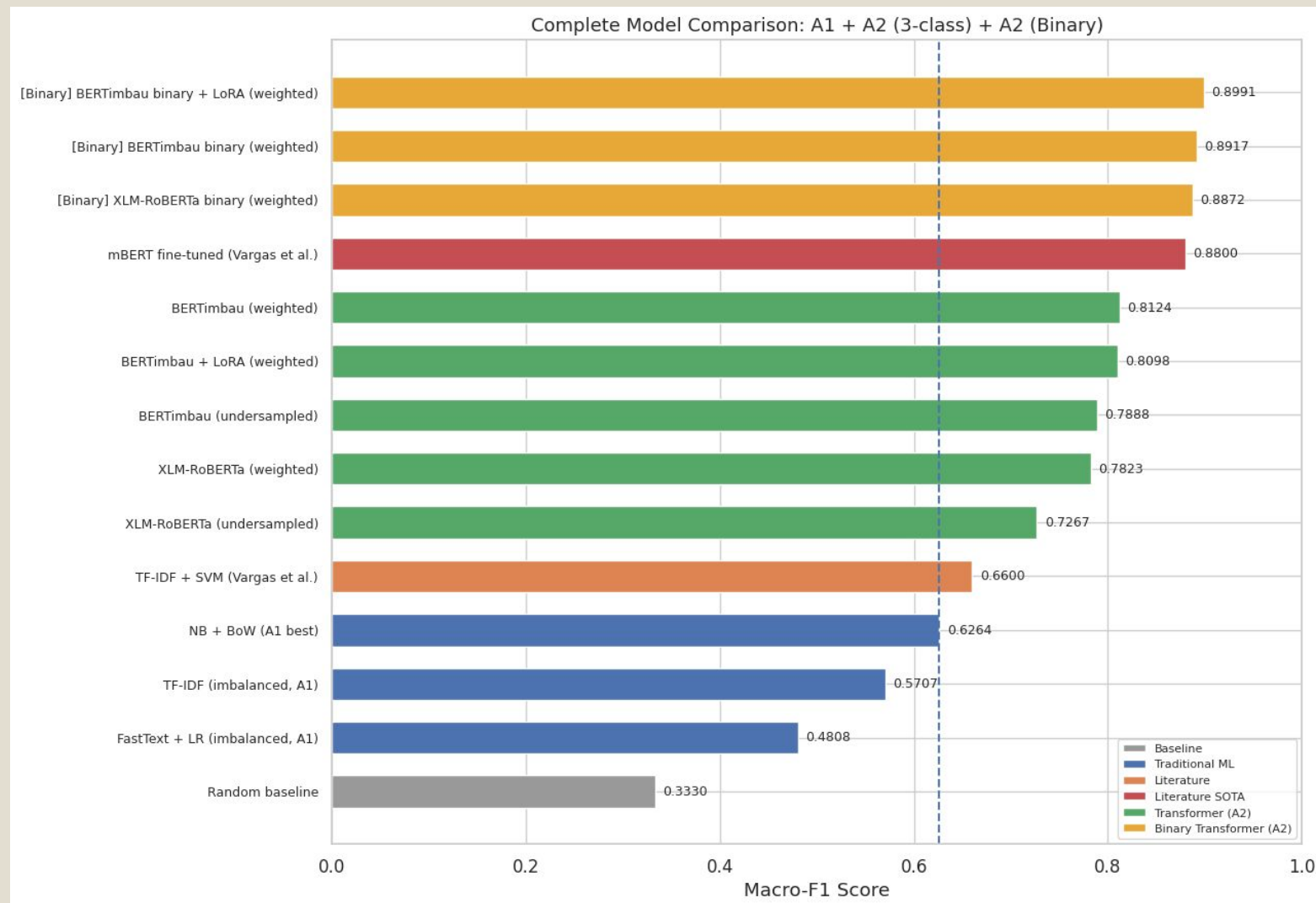


Key Insights

- **BERTimbau (weighted) leads:** Monolingual portuguese pre-training wins on this task
- **Weighted loss > undersampling:** Opposite to A1 → transformers benefit from all data with upweighted minority loss
- **LoRa is competitive:** Comparable Macro-F1 with ~98% fewer trainable parameters

Binary Experiment: Non-Factual vs Factual

- ? **Research question:** If Bias $\leftrightarrow$ Factual is the primary confusion, does collapsing Bias + Quote $\rightarrow$ Non-Factual improve boundary detection?
- Mapping:** Quote(0) + Bias(2) $\rightarrow$ Non-Factual(0) · Factual(1) $\rightarrow$ Factual(1)
- New balance:** 31.5% Non-Factual vs 68.5% Factual

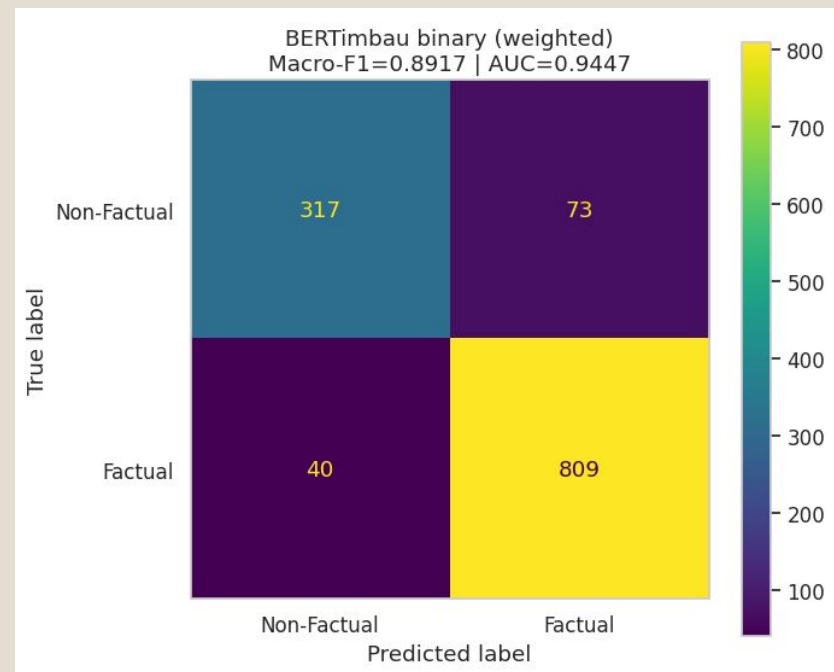


Key Insights

- **Binary Macro-F1 is higher**, simplified boundary is easier to learn
- **Factual class F1** improves with binary mapping
- **Limitation:** Binary output cannot distinguish Quote errors from Bias errors — loses interpretability for fact-checking pipelines

Error Analysis — Where Do Transformers Still Fail?

Confusion Matrix - Best Tuned Model (2 and 3-Class)



Original/Binary: Factual | Predicted: Non-Factual

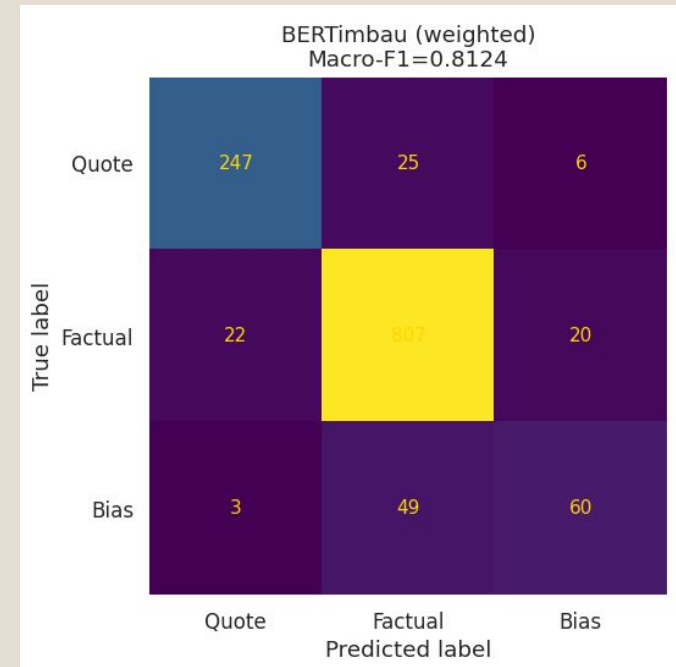
Input: Após boa trama, Robinho recebeu na entrada da área e finalizou rasteiro, porém torto, à esquerda da meta defendida por Viteri.

Hypothesis: Factual sentence contains evaluative or speech-like phrasing that the binary model associates with Non-Factual

Original: Bias | Binary: Non-Factual | Predicted: Factual

Input: Até o ano passado, o trabalho de fiscalização sobre as declarações suspeitas era mais burocrático e demorado.

Hypothesis: Bias sentence lacks strong surface markers, so the model cannot distinguish subtle non-factual language from reporting.



Original: Bias | Predicted: Factual

Input: Reajuste salarial de 91% no Congresso revolta o país.

Hypothesis: Subtle bias, since the sentence may state facts with implicit opinion framing

Original: Quote | Predicted: Factual

Input: Ele diz que vai cumprir a "quarentena" e não vai aceitar nem sondagens no setor de aviação civil.

Hypothesis: Indirect speech, no explicit quotation marks or attribution verbs

Bias → Factual (Dominant Error)

Subjective sentences predicted as Factual. Subtle bias closely resembles neutral reporting, where even human annotators disagree on these cases.

Quote → Factual

Indirect speech without explicit attribution markers. Preprocessing stripped quotation signals that NB+BoW relied on.

t-SNE Observation (A1)

Quote forms a distinct cluster; Factual/Bias overlap significantly in semantic space — this geometric reality persists even with contextual embeddings.



Bias class remains the performance bottleneck across all models in both assignments. The Bias→Factual confusion is a fundamental linguistic ambiguity, not just a modelling limitation.

Conclusions

1

Transformers Outperform Traditional ML

Best transformer Macro-F1 significantly exceeds NB+BoW (0.626). Contextual embeddings capture word order, syntax, and semantics — all lost in BoW.

2

BERTimbau > XLM-RoBERTa

Monolingual Portuguese pre-training wins. Dedicated vocabulary and attention patterns optimised for Brazilian Portuguese morphology outperform multilingual generalization.

3

Weighted Loss > Undersampling (for Transformers)

Opposite to A1: transformers benefit from all training data with upweighted minority loss. More data wins when the model learns contextual patterns.

4

LoRA is Viable

~2.5% of full fine-tuning parameters · comparable Macro-F1 · significant memory and training time savings. Practical for resource-constrained fine-tuning.

Limitations

- Bias class (9%) remains bottleneck across all approaches
- Small dataset size (taking into account the amount of bias and quote sentences)
- Binary collapsing loses Quote/Bias interpretability

Future Work

- BERTimbau-large · Ensemble methods · Text augmentation for Bias class
- Document-level context features (sentence position in article)
- Sentence-BERT for dense embeddings comparison